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Agent Systems

Agent Definition and Interaction Modelling

Project

Food chain with 10% Energy Rule

The aim of this model is to illustrate population fluctuations based on the ten percent law of energy transfer, as well as to present the survival probability of the fifth trophic level within a simple food chain

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Formal Definition and Classification of Agent Types

In the proposed Agent-Based Model (ABM), agents are defined as autonomous, discrete entities representing individual biological organisms situated within a two-dimensional spatial environment. The fundamental behavior of all active agents in this system is governed by the thermodynamic necessity to acquire, expend, and transfer energy according to the ten percent law of ecology.

To accurately model the trophic cascade and observe the survival probability of apex predators, the biological agents are classified strictly by their trophic level. This classification dictates their feeding relationships, metabolic constraints, and spatial behaviors.

Classification of Agents:

- **Level 1 Agents (Producers / Autotrophs):** These agents constitute the foundational energetic base of the ecosystem (e.g., vegetation or primary biomass). Unlike consumer agents, Level 1 agents do not exhibit hunting behaviors. They either spawn continuously at a predetermined background rate or generate energy autonomously up to a carrying capacity. They serve as the exclusive energy source for Level 2 agents.
- **Level 2 Agents (Primary Consumers / Herbivores):** These agents possess active behaviors for foraging and consuming Level 1 agents. They represent the first stage of systemic energetic loss, converting only 10% of the consumed Level 1 energy into their own biological and reproductive reserves.
- **Level 3 Agents (Secondary Consumers / Small Carnivores):** These predatory agents are programmed to hunt Level 2 agents exclusively. Because of the compounding 90% energy loss at each preceding step, their survival requires a higher rate of successful predation across a wider spatial area.
- **Level 4 Agents (Tertiary Consumers / Large Carnivores):** These agents represent significant predators that consume Level 3 agents. Their population sizes are inherently restricted by the severely diminished aggregate energy available at this tier of the food web.
- **Level 5 Agents (Quaternary Consumers / Apex Predators):** Representing the pinnacle of the modeled food chain, these agents prey exclusively on Level 4 agents. The primary analytical objective of this simulation is tracking the viability and survival probability of this specific agent class, as sustaining even a minimal population of Level 5 agents requires a massive, hyper-stable energetic foundation from Levels 1 through 4.

The Spatial Environment (Passive Agent): Furthermore, the simulation utilizes the spatial grid (the environment) as a passive, non-biological agent. The environment holds vital state variables—such as grid coordinate occupancy and environmental carrying capacity—and facilitates the spatial interactions and collision detection required for consumption events to occur between the biological agents.

Outline of Key Characteristics (State Variables)

To govern the lifecycle and interactions of the biological agents, each agent maintains a specific set of state variables. These characteristics determine their energy dynamics and survival in accordance with the 10% energy transfer rule:

- **Current Energy:** The primary state variable. It depletes continuously over time and replenishes upon successful feeding.
- **Maximum Energy Capacity:** The biological limit of energy an agent can store, scaled by its specific trophic level.
- **Metabolic Rate:** A fixed amount of energy deducted per simulation step (tick) to represent the baseline biological cost of survival and movement.
- **Energy Transfer Yield:** Fixed at exactly 10% of the consumed prey's energy. This hardcoded variable enforces the fundamental thermodynamic loss between trophic levels.
- **Starvation Threshold:** A conditional trigger; if *Current Energy* < 0, the agent is immediately removed from the simulation (mortality).
- **Reproduction Threshold:** An energy benchmark that, when exceeded, allows the agent to spawn a new agent of the same class, subsequently reducing the parent's *Current Energy* by a predetermined cost.
- **Perception/Movement Radius:** The spatial range within the grid that an agent can scan to detect adjacent prey or evade predators.

Relationships Between Agents

The fundamental interactions within this simulated ecosystem are defined by trophic predation, spatial proximity, and indirect competition. There are no symbiotic or cooperative behaviors programmed into this model; rather, the system is driven entirely by the energetic necessities outlined by the ten percent law. The relationships can be categorized into three primary dynamics:

1. Hierarchical Trophic Interactions (Predator-Prey Dynamics)

The core relationship between biological agents is one of strict, unidirectional consumption.

- **Strict Stratification:** An agent of trophic level n is programmed to recognize and consume only agents of level $n-1$. For example, Level 3 agents hunt Level 2 agents and are completely blind to (or unable to consume) Level 1 or Level 4 agents.
- **Unidirectional Energy Flow:** Energy flows exclusively upward through the trophic levels. This creates a "bottom-up" controlled ecosystem, meaning any perturbation or collapse at the producer level (Level 1) will sequentially impact all subsequent relationships up the chain.

2. Spatial Interaction Constraints

In an ABM, relationships are not global; they are localized and geographically constrained.

- **Proximity Requirement:** For a predatory relationship to result in an energetic transfer (a "consumption event"), the predator and prey agents must occupy the same spatial coordinate (or an immediately adjacent cell, depending on the grid mechanics).
- **Hunt and Evasion Mechanics:** The relationship is antagonistic. When within a defined sensory radius, predators actively attempt to close the spatial distance to their prey, while prey agents actively attempt to maximize the distance between themselves and predators.

3. Systemic Energetic Dependency (The 10% Rule)

The defining characteristic of relationships in this model is the severe energetic penalty imposed by the ten percent law of energy transfer.

- **Trophic Fragility:** Because 90% of energy is lost as heat or metabolic waste at each step, higher-tier agents require exponentially larger populations of lower-tier agents to sustain a single individual.
- **Apex Predator Vulnerability:** The relationship between Level 5 agents and the rest of the ecosystem is highly asymmetrical. The Level 5 agents are completely reliant on the systemic stability of the lower four levels. Even minor fluctuations in the Level 1 or Level 2 populations can create an energetic deficit that causes immediate starvation and localized extinction of the Level 5 apex predators.

4. Intraspecific Competition

While agents do not actively hunt members of their own trophic tier, they share an antagonistic relationship via indirect competition.

- **Resource Depletion:** Agents of the same classification compete for the same localized, finite pool of prey. High density of a specific predator in one area will rapidly deplete the local prey base, introducing self-regulating population limits (carrying capacity) within each specific trophic tier.

Expected Emergent Behaviors

By implementing simple biological and thermodynamic rules at the individual agent level, we anticipate several complex, emergent phenomena to manifest within the macro-ecosystem:

1. **Lotka-Volterra Cycles:** Natural, dynamic fluctuations in population sizes across trophic levels. The apex predator population will experience extreme volatility and potential extinction events due to energy scarcity.
2. **Spatial Clustering:** Lower-tier consumers may spontaneously form herds or clusters. This clustering reduces the mathematical probability of individual predation, forcing predators to adapt their approach.
3. **Optimized Foraging Paths:** Without hardcoded pathfinding scripts, top-tier predators are expected to develop complex, energy-efficient movement patterns to avoid starvation in barren zones.

Technical Framework

The simulation will be engineered entirely from scratch utilizing **Kotlin Multiplatform (KMP)**. This approach was selected to prioritize modern software architecture, type safety, high performance, and cross-platform accessibility, allowing the simulation to be compiled and executed natively across Desktop (JVM), Web (WebAssembly/JS), or Mobile environments without duplicating core biological logic.

The technical implementation will rely on the following architectural components:

- **Core Language (Kotlin):** The simulation engine, including agent state management, grid spatial calculations, and the discrete-time "tick" system, will be written in pure Kotlin. This ensures the ecological logic (the 10% energy transfer rule) is completely decoupled from any platform-specific UI or hardware constraints.
- **Concurrency and State Management:** The system will leverage *Kotlin Coroutines* to manage the simulation loop efficiently. Agent states and population metrics will be exposed via *StateFlow*, allowing the user interface to reactively observe and render the ecosystem's fluctuations in real-time without blocking the simulation thread.
- **Visualization and UI:** Graphical rendering of the spatial grid and real-time population charting will be implemented using *Compose Multiplatform*. This declarative UI framework allows for the creation of a unified, interactive visualizer that shares a single codebase across the target deployment platforms.